Stroma World Spec — §5: Rules of the World

> **What this section is:** The complete law book of the simulation. If a behavior isn't described by a rule here, it doesn't happen. **Format:** Plain English. Numbered. No equations. Each rule says what happens, when it happens, and why it matters. **Scope tagging:** Rules are tagged as **[ACTIVE v1.0]** (enforced in the current simulation) or **[DEFERRED]** (written now for completeness, enforced in a later cycle when the relevant actors arrive). Deferred rules are included because the v1.0 architecture must not contradict them — we're building a house that these rules will move into later.

Part A: Tumor Cell Rules

These govern what tumor cells do, how they change, and what limits them.

A1. Proliferation

Rule 1. A tumor cell attempts to divide at a rate determined by its genotype. Cells with KRAS=GOF and MYC=GOF and intact growth signaling divide fast. Cells with weaker growth drive divide slower. The rate is set once per time step from the combination rules and the calibration knobs, not negotiated or chosen by the cell. **[ACTIVE v1.0]**

Rule 2. A tumor cell cannot divide if it is under severe mechanical compression. When surrounding ECM and neighboring cells exert enough pressure on a cell, it enters growth arrest regardless of its genotype. The pressure threshold is the same for all tumor cells. This is not a checkpoint — it is a physical constraint. A cell crushed against a rigid wall does not split, even if every genetic brake is broken. **[ACTIVE v1.0]**

Rule 3. A tumor cell that has undergone EMT (ZEB1=ON) divides more slowly than an equivalent epithelial cell. A cell that is migrating is not simultaneously dividing at full speed. This is the go-versus-grow tradeoff. It is a penalty, not a block — mesenchymal cells still divide, just less often. **[ACTIVE v1.0]**

Rule 4. When a tumor cell divides, the daughter cell inherits the parent's full genotype (all 21 gene states), occupies an adjacent position, and begins life with all Boolean activity states reset to OFF. The daughter must sense its own local environment and activate its own HIF1A, ZEB1, NRF2, and ABCB1 from scratch. A daughter born in a hypoxic zone will quickly activate HIF1A on its own, but it is not born pre-adapted. **[ACTIVE v1.0]**

Rule 5. When a tumor cell divides, the daughter cell has zero intracellular drug and a fresh drug-exposure clock (NULL). Efflux resistance is not inherited — it must be independently induced by drug exposure. A resistant parent does not produce a resistant daughter unless the daughter is also exposed to drug. **[ACTIVE v1.0]**

A2. Death

Rule 6. A tumor cell can die by apoptosis. The probability of death per time step is low at baseline (set by the apoptosis resistance knob) and increases when intracellular drug concentration is high. A cell with high baseline resistance and active efflux pumps is very hard to kill. A cell with high baseline resistance but no drug exposure effectively never dies spontaneously. **[ACTIVE v1.0]**

Rule 7. When a tumor cell dies, it is removed from the simulation. It stops consuming oxygen, stops secreting signals, and stops exerting mechanical pressure. The space it occupied becomes available. Dead cells do not leave behind debris, necrotic signals, or damage-associated molecular patterns in v1.0. This is a simplification — real necrosis triggers inflammation, but there are no immune cells to respond to it yet. **[ACTIVE v1.0]**

Rule 8. A tumor cell that has never been exposed to drug is more vulnerable to drug than a cell that has been exposed and survived. First exposure is the most dangerous moment. Every hour of surviving drug exposure brings the cell closer to activating efflux, after which the drug is progressively neutralized. This creates a population-level pattern: early treatment cycles kill more cells than later cycles, because the survivors are increasingly hardened. **[ACTIVE v1.0]**

A3. Mutation and Adaptation

Rule 9. Gene states do not change during a simulation run. A cell born with KRAS=GOF dies with KRAS=GOF. There is no point mutation, no gene amplification, no epigenetic silencing during the run. The genotype is frozen at initialization. This means there is no clonal evolution in v1.0 — every tumor cell in a run shares the same 21-gene genotype, and there are no resistant subclones with novel mutations. **[ACTIVE v1.0]**

Rule 10. Phenotypic adaptation does occur, but it is entirely signal-driven and reversible. A cell in a hypoxic zone activates HIF1A and becomes drug-resistant and secretion-amplified. If that same cell moves (or is pushed) to a well-oxygenated zone, HIF1A deactivates and those phenotype changes reverse. Nothing permanent happened to the cell's genotype. It adapted to its environment, not to its evolutionary history. **[ACTIVE v1.0]**

Rule 11. Drug-induced efflux activation (NRF2 → ABCB1) is the one adaptation that has a persistent memory component. Once ABCB1 activates, it does not instantly deactivate when drug is removed. NRF2 decays slowly, and ABCB1 remains active for a period after drug withdrawal. This creates a population of "recently exposed, still resistant" cells that are temporarily protected even after treatment ends. The decay is slow enough that a second drug pulse arriving shortly after the first will face partial resistance, but long enough drug-free intervals allow resistance to fade. **[ACTIVE v1.0]**

Rule 12. [DEFERRED — Cycle 3: Clonal Evolution] When stochastic mutation is enabled, a dividing tumor cell has a small probability of altering one gene state in its daughter (WT→LOF, WT→GOF, or reversal). This creates intra-tumor heterogeneity over time and allows resistant subclones to emerge de novo. The v1.0 architecture supports this by tracking gene states per-cell rather than per-population, but the mutation event itself is not triggered. **[DEFERRED]**

A4. Secretion

Rule 13. Every living tumor cell secretes TGF-β and SHH into its local microenvironment at every time step. This is constitutive — it happens regardless of oxygen levels, drug exposure, or barrier state. The tumor does not choose to build its stroma; it cannot stop building it. This is a consequence of KRAS=GOF driving continuous paracrine output. **[ACTIVE v1.0]**

Rule 14. When a tumor cell's HIF1A is active (hypoxic zone), its TGF-β secretion rate increases. SHH secretion does not change under hypoxia. This means the deeper the tumor is inside its own barrier (more hypoxic), the louder it screams for more stroma — a positive feedback loop. **[ACTIVE v1.0]**

Rule 15. Secreted TGF-β and SHH diffuse outward from the cell through the microenvironment. Their diffusion is slowed by ECM density. In dense stroma, these signals don't travel as far — they concentrate near their source. This means the strongest activation signal is at the tumor-stroma interface, not at the domain edge. **[ACTIVE v1.0]**

A5. EMT and Motility

Rule 16. A tumor cell undergoes epithelial-to-mesenchymal transition when the combined EMT-inducing signal at its position exceeds the induction threshold. The signal comes from two sources: local TGF-β (acting through the SMAD-independent arm, which works regardless of SMAD4 status) and HIF1A activation (from hypoxia). Either source alone can trigger EMT if strong enough. Both together make it easier. **[ACTIVE v1.0]**

Rule 17. EMT is reversible. If a mesenchymal cell moves to a region with low TGF-β and adequate oxygen, the induction signal drops below threshold and the cell reverts to epithelial. This is why EMT is spatially restricted to the tumor-stroma interface and hypoxic zones — it is maintained by the local environment, not locked in by a permanent cell decision. **[ACTIVE v1.0]**

Rule 18. A mesenchymal tumor cell (ZEB1=ON, CDH1=OFF) moves faster, sticks less to other cells, degrades local ECM via MMP2, and upregulates PD-L1. These changes are proportional to the EMT extent tier (LOW, MEDIUM, HIGH). An epithelial tumor cell (ZEB1=OFF, CDH1=ON) barely moves, adheres strongly to its neighbors, does not degrade ECM, and has baseline PD-L1. **[ACTIVE v1.0, except PD-L1 has no functional target until Cycle 2]**

Rule 19. MMP2-mediated ECM degradation by mesenchymal tumor cells is local. It reduces ECM density only in the voxels immediately surrounding the cell. It does not dissolve the barrier globally — it carves a path at the invasive front. This local degradation is the tumor's limited ability to push through its own wall. **[ACTIVE v1.0]**

A6. TGF-β Sensitivity (The SMAD4 Asymmetry)

Rule 20. This is the single most important rule in the simulation. TGF-β has two arms of signaling, and SMAD4 status determines which arms are active in a given cell:

- **SMAD-dependent arm (braking):** TGF-β induces growth arrest and apoptosis. This arm requires SMAD4. In SMAD4=WT cells, TGF-β slows proliferation and can trigger death. In SMAD4=LOF cells (canonical PDAC), this arm is severed. TGF-β cannot slow these cells.
- **SMAD-independent arm (invasion):** TGF-β drives EMT, motility, and invasion via MAPK/RhoA pathways. This arm does NOT require SMAD4. It works in all tumor cells regardless of SMAD4 status.

The result: in canonical PDAC (SMAD4=LOF), TGF-β drives invasion and EMT in the tumor while simultaneously activating the stroma to build more barrier. The brake is severed but the gas pedal works. The tumor floods its environment with a signal that builds a wall around it and makes it more invasive — and it cannot be stopped by that same signal. **[ACTIVE v1.0]**

Rule 21. Stromal cells always have SMAD4=WT. Both arms of TGF-β signaling are active in stroma. TGF-β fully activates stellate cells, drives ECM production, and promotes CAF survival. There is no SMAD4 asymmetry in the stroma. The asymmetry is exclusively a tumor cell property. **[ACTIVE v1.0]**

Part B: Stromal Cell Rules

These govern how quiescent stellate cells become barrier-building fibroblasts and what happens after.

B1. Activation

Rule 22. A quiescent pancreatic stellate cell (PSC) becomes an activated cancer-associated fibroblast (CAF) when the combined TGF-β and SHH signal at its position exceeds the activation threshold. Either

signal alone can trigger activation if strong enough. Both together make activation faster and more certain. **[ACTIVE v1.0]**

Rule 23. Activation is irreversible in v1.0. Once ACTA2 turns ON, it stays ON for the life of the cell. The PSC is now a CAF. It will produce ECM, secrete TGF-β, proliferate, and contract its surroundings until it dies (which, in v1.0, is approximately never). This is a simplification — real CAFs have some plasticity — but it prevents oscillatory activation-deactivation cycles that would complicate early debugging without changing the qualitative barrier outcome. **[ACTIVE v1.0]**

Rule 24. [DEFERRED — Cycle 2: CAF Subtypes] When subtype heterogeneity is enabled, activated CAFs can exist in multiple modes (myofibroblastic, inflammatory, antigen-presenting) with different behaviors. The v1.0 architecture treats all CAFs as one type approximating myofibroblastic behavior (ECM-producing, contractile). **[DEFERRED]**

B2. Barrier Construction

Rule 25. An activated CAF produces extracellular matrix at its position at every time step. The ECM it produces has two components: hyaluronic acid (HA, from HAS2) and fibrillar collagen (from COL1A1). Both components are deposited into the same ECM density field at the cell's location, but their relative fraction is tracked because they affect physics differently. **[ACTIVE v1.0]**

Rule 26. If the CAF's GLI1 is also active (SHH pathway receiving signal), ECM production is boosted above the baseline activated rate. GLI1 is the SHH-responsive amplifier: it makes an already-active CAF produce even more matrix. If SHH signal disappears (e.g., SHH inhibition), GLI1 turns off and production drops back to the baseline activated rate — but does not stop, because TGF-β/ACTA2-driven production continues independently. This is why SHH inhibition only partially thins the barrier. **[ACTIVE v1.0]**

Rule 27. An activated CAF secretes TGF-β into the microenvironment. This is the positive feedback loop: tumor secretes TGF-β → activates PSCs → CAFs secrete more TGF-β → activates more PSCs → more CAFs → more TGF-β. This loop is what makes the barrier self-amplifying rather than self-limiting. Without CAF TGF-β feedback, the barrier would only grow as far as the tumor's own secretion can reach. With it, the stroma recruits itself. **[ACTIVE v1.0]**

Rule 28. ECM does not diffuse. It stays where it was deposited. Collagen fibers and HA gel are structural — they do not flow to neighboring voxels. This means the barrier's spatial structure is entirely determined by where CAFs are and how long they've been producing. Dense ECM near the tumor, sparse ECM far from it, because that's where the activated CAFs are. **[ACTIVE v1.0]**

Rule 29. ECM has a very slow natural turnover rate. In the absence of active degradation (MMP2 or exogenous intervention), ECM density at a voxel decreases extremely slowly over time. For practical purposes in the v1.0 simulation timescale (weeks), un-degraded ECM is nearly permanent. This is why the barrier persists after drug treatment: the drug kills tumor cells but does not dissolve the ECM, and surviving CAFs continue maintaining it. **[ACTIVE v1.0]**

B3. CAF Proliferation and Persistence

Rule 30. Activated CAFs divide at a moderate rate — substantially slower than tumor cells. CAF proliferation is the mechanism by which the fibroblast population expands to fill the peri-tumoral space. It is boosted when GLI1 is active (SHH signaling). Quiescent PSCs barely divide at all. **[ACTIVE v1.0]**

Rule 31. CAFs do not respond to the drug substrate. Standard PDAC chemotherapy targets tumor cells, not fibroblasts. CAFs are drug-indifferent: drug diffuses through them, past them, and they persist regardless. This is why the barrier rebuilds after treatment. The drug may kill the tumor cells that initiated the barrier, but the CAFs that maintain it are untouched. When surviving tumor cells resume secreting, the intact CAF population is ready to resume production immediately. **[ACTIVE v1.0]**

Rule 32. CAFs have a very low spontaneous death rate — effectively zero in v1.0. Once activated, they persist for the duration of the simulation. This is an intentional simplification that reflects the clinical reality: PDAC stroma is durable. Tumors resected after neoadjuvant chemotherapy still show dense, viable stroma even when the tumor cellularity is reduced. **[ACTIVE v1.0]**

Part C: Barrier Physics Rules

These govern how the ECM barrier interacts with everything that tries to move through it.

C1. Diffusion Impedance

Rule 33. ECM density reduces the diffusion rate of all substrates passing through it. Higher ECM means slower diffusion of oxygen, TGF-β, SHH, and drug. The relationship is monotonic: more ECM always means slower diffusion, never faster. This is the fundamental mechanism by which the barrier blocks drug delivery and creates hypoxia. **[ACTIVE v1.0]**

Rule 34. The HA component of ECM impedes diffusion more than the collagen component for small molecules and drug. HA creates a hydrated gel that acts as a sieve — it generates interstitial fluid pressure that opposes convective transport and slows diffusive transport through gel resistance. Collagen provides structural scaffolding but has wider pore spaces between fibers. This is why HA depletion (hyaluronidase) improves drug delivery more than collagen depletion alone in the model. **[ACTIVE v1.0]**

Rule 35. The collagen component of ECM provides mechanical stiffness more than the HA component. Collagen fibers resist compression and deformation. HA gel is soft and compressible under pressure. This is why collagen depletion releases mechanical confinement (tumor expands) more than HA depletion. **[ACTIVE v1.0]**

Rule 36. ECM also slows the diffusion of the paracrine signals (TGF-β, SHH) that created it. Dense ECM near the tumor traps these signals locally, concentrating them at the tumor-stroma interface. This has two consequences: stromal activation is strongest right at the boundary (where it builds the most ECM), and TGF-β-driven EMT is strongest at the same boundary (where tumor cells are most exposed). The barrier reinforces its own construction by concentrating the signals that build it. **[ACTIVE v1.0]**

C2. Mechanical Confinement

Rule 37. A growing tumor pushes against surrounding ECM. If the ECM is stiff (high collagen fraction, high density), it pushes back. The resulting solid stress slows or halts tumor cell division at the boundary through contact inhibition. The tumor interior, where cells are pushing outward but are buffered by other tumor cells, experiences less direct ECM contact and can continue dividing — but the boundary cells are pressed against the wall. **[ACTIVE v1.0]**

Rule 38. Proliferation-driven pressure from the tumor passively compresses adjacent ECM, increasing its density even without new CAF production. This is a mechanical effect: expansion squeezes the matrix tighter. The compaction is proportional to the expansion force and the pre-existing ECM stiffness. This is the non-biochemical arm of barrier densification — smaller than CAF-driven production, but it contributes to interstitial pressure and is the reason even stroma without active CAFs can impede drug delivery to some degree. **[ACTIVE v1.0]**

Rule 39. When stroma is removed (by degradation or SHH inhibition), the mechanical confinement is released. Tumor cells at the boundary are no longer compressed. They resume dividing and expanding into the freed space. If no drug is present to kill them in this newly accessible zone, the tumor grows larger than it would have with the stroma intact. This is why stroma removal without concurrent cytotoxicity can worsen outcomes. The stroma was both a shield and a cage. **[ACTIVE v1.0]**

C3. Effects on Infiltration

Rule 40. [DEFERRED — Cycle 2: Immune Actors] When immune cells are present, ECM density impedes their migration just as it impedes drug diffusion. A T cell trying to move through dense stroma is slowed in proportion to ECM density. In very dense stroma, T cell migration effectively stalls — the cells cannot physically reach the tumor. This creates the "immune exclusion" phenotype: T cells may be present at the stroma periphery but fail to penetrate to the tumor core. The v1.0 architecture already tracks ECM density with sufficient spatial resolution to support this rule — it is simply not yet triggered because there are no T cell actors. **[DEFERRED]**

Rule 41. [DEFERRED — Cycle 2] Collagen fiber alignment in the ECM can either impede or guide immune cell migration. Fibers oriented perpendicular to the direction of immune approach create a barrier. Fibers oriented parallel create a highway. In v1.0, ECM is isotropic (no fiber orientation tracked), so this rule cannot be enforced. When fiber alignment is added as a variable on the ECM field, this rule activates and creates a more realistic infiltration landscape. **[DEFERRED]**

Rule 42. [DEFERRED — Cycle 2] Even when immune cells successfully penetrate the stroma, the density of ECM they passed through slows their arrival. Drugs and immune cells racing to reach the tumor interior face the same physical obstacle. The barrier buys the tumor time — time to adapt (Rule

11), time to undergo EMT and upregulate PD-L1 (Rule 18), and time to establish the hypoxic immunosuppressive microenvironment (Part E). **[DEFERRED]**

Part D: Immune Recognition Rules

These describe how the immune system identifies (or fails to identify) tumor cells. All deferred to Cycle 2 but architecturally planned.

D1. Recognition

Rule 43. [DEFERRED — Cycle 2] Immune recognition of tumor cells is uncertain. A T cell that reaches a tumor cell does not automatically kill it. Recognition depends on: (a) whether the tumor cell presents recognizable antigens (related to mutation burden, not modeled in detail in v1.0's fixed genotype), and (b) whether recognition is blocked by checkpoint ligands (PD-L1, tracked via ZEB1 state in v1.0 but with no functional immune target). The probability of recognition is less than one even under ideal conditions. **[DEFERRED]**

Rule 44. [DEFERRED — Cycle 2] Mesenchymal tumor cells (ZEB1=ON) are harder to recognize than epithelial tumor cells. EMT upregulates PD-L1 and downregulates MHC class I presentation. This means the invasive front — the tumor surface that an immune cell encounters first after penetrating the stroma — is the most immune-evasive part of the tumor. The architecture for this is already in place: ZEB1 state is tracked per-cell (§1.3 of State Variables), and the immune-evasion tendency (IET) axis from the Mutation→Behavior Map is computed as bookkeeping. When immune actors arrive, they read these values. **[DEFERRED]**

Rule 45. [DEFERRED — Cycle 2] Hypoxic tumor cells (HIF1A=ON) are harder to kill even when recognized. Hypoxia generates an immunosuppressive microenvironment through adenosine accumulation, metabolic hostility (low glucose, high lactate), and direct suppression of T cell effector function. A T cell that reaches a hypoxic tumor cell and recognizes it still has a reduced kill probability. **[DEFERRED]**

D2. Collateral Damage

Rule 46. [DEFERRED — Cycle 2] Immune effector activity is not perfectly targeted. When a T cell kills a tumor cell, inflammatory mediators released in the process can damage nearby cells — including other tumor cells (beneficial collateral), stromal cells (stroma disruption), and normal tissue cells if they exist at the domain boundary (harmful collateral). The radius and magnitude of collateral damage are parameters that the Cycle 2 immune module will define. **[DEFERRED]**

Rule 47. [DEFERRED — Cycle 2] Immune-mediated stroma disruption is a double-edged collateral effect. Killing tumor cells at the tumor-stroma interface damages nearby CAFs and degrades local ECM (beneficial — opens the barrier). But it also releases sequestered TGF-β from degraded ECM (harmful — activates more stroma) and creates inflammatory signals that can recruit immunosuppressive cells. The net effect of immune collateral on the barrier is context-dependent and is one of the key dynamics Cycle 2 must resolve. **[DEFERRED]**

Rule 48. [DEFERRED — Cycle 2] Excessive immune activation can damage host tissue. If collateral damage at the domain boundary exceeds a threshold, it represents autoimmune-like toxicity. The fitness function in Cycle 2 must penalize interventions that trigger immune responses so aggressive that they compromise normal tissue integrity — echoing the real clinical problem of immune-related adverse events from checkpoint inhibitors. **[DEFERRED]**

Part E: Suppression Rules

These govern how the tumor and its stroma suppress immune and therapeutic attack.

E1. Stroma-Mediated Suppression

Rule 49. Dense ECM physically excludes immune cells (Rule 40) and blocks drug delivery (Rule 33). These are the primary suppression mechanisms in v1.0. They are passive — the stroma doesn't "decide" to exclude anything, it simply exists as a physical obstruction. The suppression is proportional to barrier

density: thicker stroma means more exclusion, thinner stroma means less. This is already fully functional in v1.0 for drug exclusion and tracked (via ECM density) for future immune exclusion. **[ACTIVE v1.0 for drug; DEFERRED for immune cells]**

Rule 50. [DEFERRED — Cycle 2] CAFs actively suppress immune function beyond physical exclusion. Activated CAFs secrete immunosuppressive factors (TGF-β, IL-6, CXCL12) that directly inhibit T cell proliferation and effector function even if T cells reach the tumor. This means that a thin stroma is not automatically immune-permissive — even partial barrier penetration exposes immune cells to a biochemically hostile environment. CAF-mediated immune suppression is a function of CAF density and activation state, both of which are already tracked in v1.0. **[DEFERRED]**

E2. Tumor-Intrinsic Suppression

Rule 51. Tumor cells suppress drug efficacy through the efflux mechanism (Rules 8, 11). A drug that penetrates the barrier and reaches the tumor still faces resistance from ABCB1-mediated efflux. This is the second line of defense after the physical barrier. The stroma blocks the drug; if the drug gets through, the tumor pumps it out. **[ACTIVE v1.0]**

Rule 52. [DEFERRED — Cycle 2] Tumor cells suppress immune efficacy through checkpoint ligand expression. PD-L1 (upregulated by ZEB1 in mesenchymal cells and by HIF1A in hypoxic cells) binds PD-1 on T cells and inhibits the kill signal. The cells most exposed to immune attack (peripheral, invasive front) are the most PD-L1-positive (because they're the most mesenchymal). This is tracked in v1.0 via ZEB1 and HIF1A states but has no functional consequence until immune actors arrive. **[DEFERRED]**

Rule 53. [DEFERRED — Cycle 2] The tumor microenvironment recruits immunosuppressive cells. KRAS-driven GM-CSF secretion recruits myeloid-derived suppressor cells. TGF-β induces regulatory T cells. These populations actively suppress cytotoxic immune responses. The v1.0 paracrine output (TGF-β) is already tracked as a microenvironment field and could serve as the input signal for suppressive cell recruitment when those actors are added. **[DEFERRED]**

E3. Layered Defense Model

Rule 54. The tumor's defenses are layered, not redundant. Each layer must be breached in order, and each layer becomes the relevant defense only when the outer layers fail:

- **Layer 1 — Physical barrier (stroma).** Blocks drug delivery and immune infiltration. If this layer holds, nothing reaches the tumor and the inner layers are irrelevant.
- **Layer 2 — Biochemical suppression (checkpoint ligands, immunosuppressive milieu).** If something penetrates the stroma, it encounters an immunosuppressive and drug-hostile environment. T cells are inhibited by PD-L1 and TGF-β. Drug is diluted by poor diffusion and metabolized in the hypoxic environment.
- **Layer 3 — Cell-intrinsic resistance (efflux, apoptosis resistance).** If drug reaches the tumor cell at sufficient concentration, the cell's own machinery (ABCB1, BCL-XL/BAX ratio, TP53 loss) determines whether it dies. The most deeply shielded cells have had the most time to activate efflux.
- **Layer 4 — Regrowth from sanctuary (proliferative capacity of survivors).** If some cells are killed but others survive (in the densest stroma, deepest from drug source), the survivors resume proliferation and paracrine secretion, rebuilding the barrier and repopulating the tumor.

An effective intervention must breach all four layers. The EA's job is to find combinations that do this. In v1.0, Layers 1, 3, and 4 are fully active. Layer 2 is partially active (drug diffusion impedance) with immune suppression deferred. **[ACTIVE v1.0: Layers 1, 3, 4. Partial: Layer 2. DEFERRED: Layer 2 immune components]**

Part F: Host-Level Rules

These govern the conditions of the whole system — what constitutes failure, success, and the limits of biological plausibility.

F1. Host Death Thresholds

Rule 55. The host (patient) dies when tumor burden exceeds a critical mass. In the simulation, this is operationalized as: if total_tumor_cells exceeds a predefined ceiling, the simulation terminates with outcome = HOST DEATH. This ceiling represents the tumor burden at which organ failure occurs — in PDAC, this is typically from biliary obstruction, vascular invasion, or metastatic organ failure, none of which are explicitly modeled, but all of which correlate with tumor mass. **[ACTIVE v1.0]**

Rule 56. The host death threshold is not the same as the domain capacity. The domain can physically hold more cells than the host can survive. A simulation that fills the domain with tumor cells has exceeded any plausible survival threshold. The host death ceiling should be set well below domain capacity — calibrated to represent a lethal tumor burden in the pancreatic context. **[ACTIVE v1.0]**

Rule 57. [DEFERRED — Cycle 2] The host can also die from treatment toxicity. If the EA evolves interventions that cause excessive collateral damage (Rule 48) or drug toxicity beyond what the host can tolerate, the fitness function penalizes this with a toxicity cost. In v1.0, toxicity is approximated by penalizing drug dose in the fitness function (higher dose = higher toxicity penalty) rather than modeling organ damage explicitly. **[ACTIVE v1.0 as fitness penalty; DEFERRED as explicit mechanism]**

Rule 58. The host death threshold creates a time pressure. The tumor is growing. The barrier is thickening. Drug efficacy is declining with each treatment cycle (Rule 8). The EA must find interventions that work fast enough to reduce tumor burden before it crosses the lethal threshold. A slow strategy that would eventually work but takes too long is a failed strategy — the host dies while the tumor is still being "slowly reduced." **[ACTIVE v1.0]**

F2. Treatment Constraints

Rule 59. Drug cannot be applied at arbitrarily high concentrations. The boundary drug concentration has a maximum value representing the maximum tolerated dose. This ceiling exists because real drugs have dose-limiting toxicities — higher doses kill more tumor cells but also cause more harm to the host. The EA is free to choose any concentration up to this ceiling but cannot exceed it. **[ACTIVE v1.0]**

Rule 60. Drug delivery is spatially uniform at the domain boundary. All boundary voxels receive the same drug concentration when treatment is applied. There is no selective delivery to one side of the tumor. This reflects the clinical reality that systemic chemotherapy distributes throughout the vasculature

— it doesn't target one face of the tumor preferentially. Spatial selectivity of drug exposure is created entirely by the barrier's differential thickness, not by delivery targeting. **[ACTIVE v1.0]**

Rule 61. The EA can schedule interventions over time. Drug can be pulsed on and off. SHH inhibition can be applied continuously or intermittently. ECM degradation can be a single event or repeated. The EA's genome encodes a temporal intervention schedule, not a single static parameter. This allows the EA to discover strategies like: "pulse drug for 6 hours, wait 18 hours for efflux to decay, pulse again" — strategies that exploit the temporal dynamics of Rules 8 and 11. **[ACTIVE v1.0]**

F3. Biological Plausibility Constraints

Rule 62. No cell can exceed a maximum biologically plausible proliferation rate. Even with all checkpoints destroyed and all growth signals active, there is a physical minimum time to replicate a genome, divide organelles, and undergo cytokinesis. This floor doubling time is approximately 20 hours for the fastest human cancer cells. The simulation clamps proliferation rate at this ceiling. **[ACTIVE v1.0]**

Rule 63. No voxel can exceed a maximum ECM density. Real tissue has a physical packing limit — you cannot compress collagen and HA into infinite density. If ECM at a voxel reaches the ceiling (normalized to 1.0), no further ECM can be deposited there. CAFs adjacent to saturated voxels continue producing, but the excess is effectively lost (or could be shunted to adjacent unsaturated voxels in a future refinement). **[ACTIVE v1.0]**

Rule 64. Cells cannot occupy the same physical space. PhysiCell's mechanics solver enforces volume exclusion — if a dividing cell tries to place its daughter in occupied space, the daughter (and surrounding cells) are pushed apart. This is the origin of the mechanical pressure that drives contact inhibition (Rule 2) and ECM compaction (Rule 38). **[ACTIVE v1.0]**

Rule 65. Oxygen concentration cannot exceed the boundary supply value anywhere in the domain interior. There is no internal oxygen source. O_2 enters from the boundary and is only consumed, never produced, by cells. Any voxel with O_2 above the boundary value is a bug. **[ACTIVE v1.0]**

Part G: Meta-Rules

Rules about how the rules work.

Rule 66. All cell decisions within a single time step are based on the environment state at the beginning of that time step. No cell reads a field that another cell has already written to in the same step. Sensing happens first, then decisions, then environment writes, then PDE solving. This prevents order-dependent artifacts and ensures deterministic reproducibility given the same random seed. **[ACTIVE v1.0]**

Rule 67. Stochastic events (death probability, motility direction noise, initial cell placement) are governed by the simulation's random seed. Same seed, same outcome. Different seeds, different stochastic trajectory but same qualitative behavior. All Reality Anchor tests require at least three replicate runs with different seeds to confirm that pass/fail results are robust to stochastic variation. **[ACTIVE v1.0]**

Rule 68. When rules conflict, physical constraints override biological signaling. If a cell's genotype says "divide" but the mechanical pressure says "no room," the cell does not divide (Rule 2 overrides Rule 1). If a cell's EMT program says "migrate through ECM" but the ECM is at maximum density, migration is slowed to near-zero (Rule 63 constrains Rule 18). Physics beats signaling. Signaling beats genotype defaults. **[ACTIVE v1.0]**

Rule 69. Any behavior not explicitly described by a rule in this section does not occur. Cells do not spontaneously change type. Fields do not spontaneously appear. Interventions do not occur unless the EA or scenario triggers them. The simulation is a closed system governed entirely by these rules, the module logic, and PhysiCell's built-in mechanics and PDE solvers. If something unexpected happens, it is either an emergent consequence of these rules (good) or a bug (bad). There is no third option. **[ACTIVE v1.0]**

Rule Index

Domain	Rules	Active in v1.0	Deferred
Tumor proliferation	1–5	5	0

Domain	Rules	Active in v1.0	Deferred
Tumor death	6–8	3	0
Mutation & adaptation	9–12	3	1 (clonal evolution)
Secretion	13–15	3	0
EMT & motility	16–19	4	0
TGF-β / SMAD4	20–21	2	0
Stromal activation	22–24	2	1 (CAF subtypes)
Barrier construction	25–29	5	0
CAF persistence	30–32	3	0
Barrier physics	33–39	7	0
Infiltration	40–42	0	3
Immune recognition	43–45	0	3
Collateral damage	46–48	0	3
Suppression	49–54	3 partial	4
Host thresholds	55–58	4	0
Treatment constraints	59–61	3	0
Biological plausibility	62–65	4	0
Meta-rules	66–69	4	0

Domain	Rules	Active in v1.0	Deferred
TOTAL	**69**	**~55 active**	**~14 deferred**

69 rules. 55 active in v1.0. 14 deferred but architecturally planned. Written in English, no equations. If the simulation does something not described here, it's a bug.